Rational Number Project

Initial Fraction Ideas Lesson 7: Overview	Materials
This lesson reinforces the idea that as the number of parts the unit is divided into increases, the size of the parts decreases.	• 8.5″ X 1″ strips of paper for each student and teacher • Student Pages A, B, C

Teaching Actions

Warm Up

Order these fraction pairs. Write an explanation for each pair (use pictures in your explanation).

$\frac{3}{4}\,\frac{3}{8}$ $\frac{5}{6}\,\frac{3}{6}$

Large Group Introduction

1. Ask children to fold a strip of paper into 2 equal parts. Using the same strip of paper ask them how they can increase the number of equal parts to 4. Have them do this, but before they open up the strip of paper to show fourths ask: Before you open up the strip, can you tell me if the size of the equal parts will be larger or smaller than halves? Why?

2. Repeat for:
   - Fourths to eighths
   - Thirds changed to sixths (Use pre-divided strips for thirds if needed)
   - Halves to sixths

3. Now ask students to fold, shade, and label these fractions with paper folding:

$\frac{1}{3}$ $\frac{1}{4}$
$\frac{1}{3}$ $\frac{1}{6}$
$\frac{3}{4}$ $\frac{1}{4}$

[Do more if needed]

Comments

Children need opportunities to use new ideas in order to ensure they internalize them.

Many experiences with physical models are needed to overcome the influence of children's whole number thinking.

In this lesson students use paper folding to reexamine the relationship between size of piece and number of pieces the whole is divided into.

Encourage children to explain their ordering. Don't let them refer to only one part of the fraction, as for example: 1/3 vs. 1/4 "thirds are bigger". Thirds may be bigger, but that information is enough to order 2 fractions only if the numerators are the same. "Thirds are bigger so 1 of a larger piece is greater than 1 of a smaller pieces." By talking like this children are coordinating numerator and denominator to approximate the size of the fraction. You want to build the notion of a fraction as a single entity!

Students may over generalize and think bigger is always more. Check for this.

Teaching Actions	Comments
Small Group/Partner Work 4. Put students in pairs and assign Student Page A. Student 1 will make fraction 1 with paper folding; student 2 will make fraction 2. They will then compare and circle the larger fraction. 5. Student Pages B and C offer extra practice. **Wrap Up** 6. Conclude the lesson by first asking children to draw a picture to solve each problem below. Have students explain their ordering strategies. Mary 1/2 of a large Domino pizza; Joan had 1/4 of a large Domino pizza. Who ate more? Lianna ate 4/8 parts of a candy bar Rodrigo ate 4/6 of a same-sized candy bar. Who ate more? 7. Now ask: Picture ¾ in your mind using the whole circle as the unit. Now picture ½ of the whole circle. Which is bigger ½ or ¾? 8. Look to see if students say ½ is greater because halves are bigger than fourths. Encourage a discussion of the role of the numerator and how 3 smaller parts may be bigger than one bigger piece.	Some children may be able to compare without manipulatives $\frac{1}{3}$ vs. $\frac{1}{5}$; $\frac{2}{10}$ vs. $\frac{2}{20}$ but there is no need to push abstraction at this level. Some students may try to compare fractions without the manipulatives and make errors. Encourage them to use paper folding at least to verify their guesses. Look for students to say something about the numerator and denominator. "Halves are bigger than fourths and you have one of each".

Translations

- Written symbols to manipulative to verbal

Order these fraction pairs. Write an explanation for each pair (use pictures in your explanation).

$\frac{3}{4}$ $\frac{3}{10}$

$\frac{5}{7}$ $\frac{3}{7}$

$\frac{1}{9}$ $\frac{1}{4}$

Name_____ Grade 3 Lesson 7/ Student Page A

Directions:
Circle the larger fraction. Use your paper strips to determine the answers.

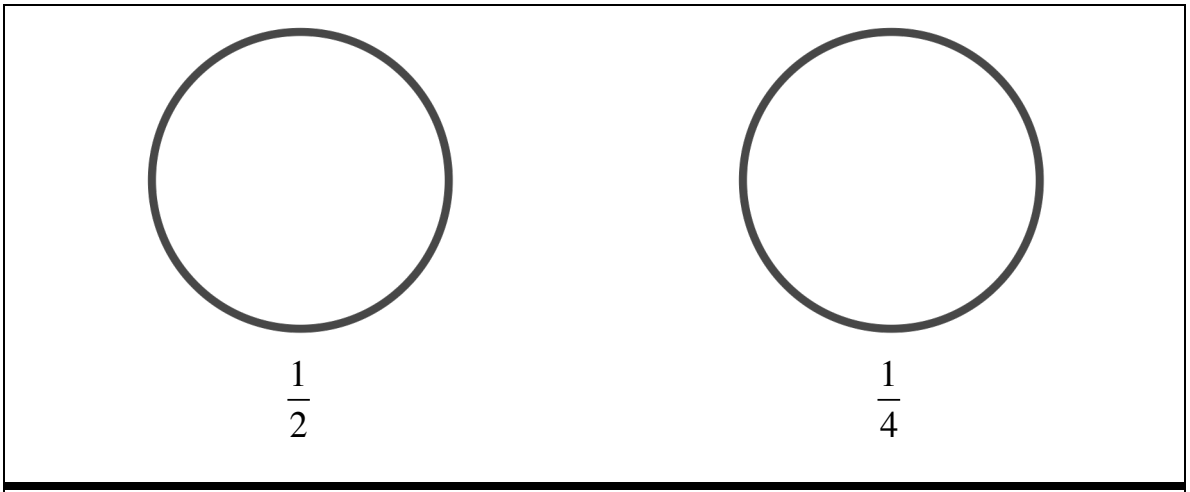
$\frac{1}{2}$	$\frac{1}{3}$	$\frac{2}{6}$	$\frac{2}{3}$
$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{2}{3}$
$\frac{3}{4}$	$\frac{1}{4}$	$\frac{3}{6}$	$\frac{3}{3}$
$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{0}{2}$
$\frac{1}{3}$	$\frac{1}{6}$		
$\frac{1}{2}$	$\frac{1}{4}$		

Without your paper strips, circle the larger fractions.

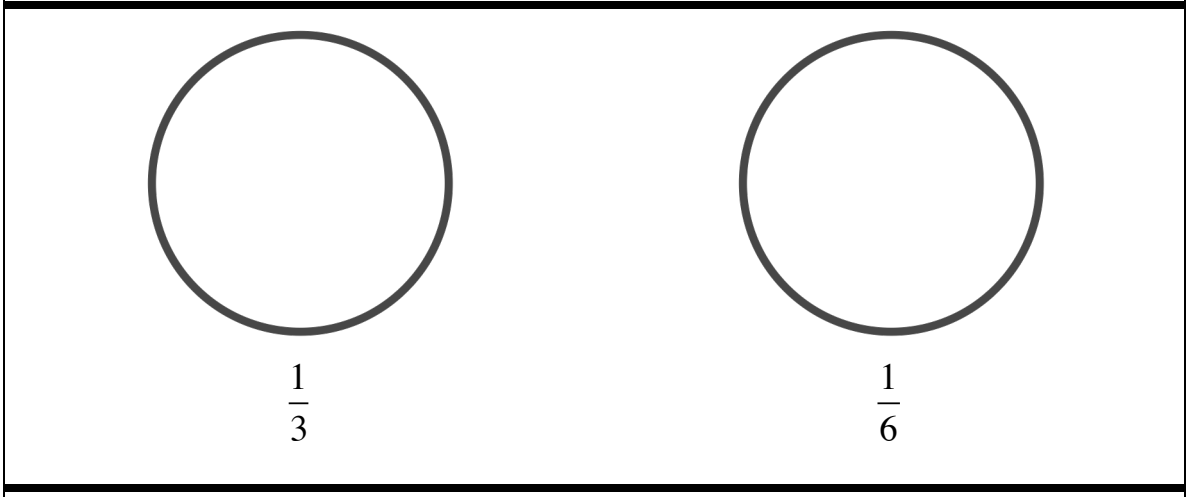
$\frac{1}{100}$	$\frac{1}{99}$
$\frac{3}{40}$	$\frac{3}{50}$
$\frac{2}{10}$	$\frac{4}{10}$

Name_____ Grade 3 Lesson 7/ Student Page B

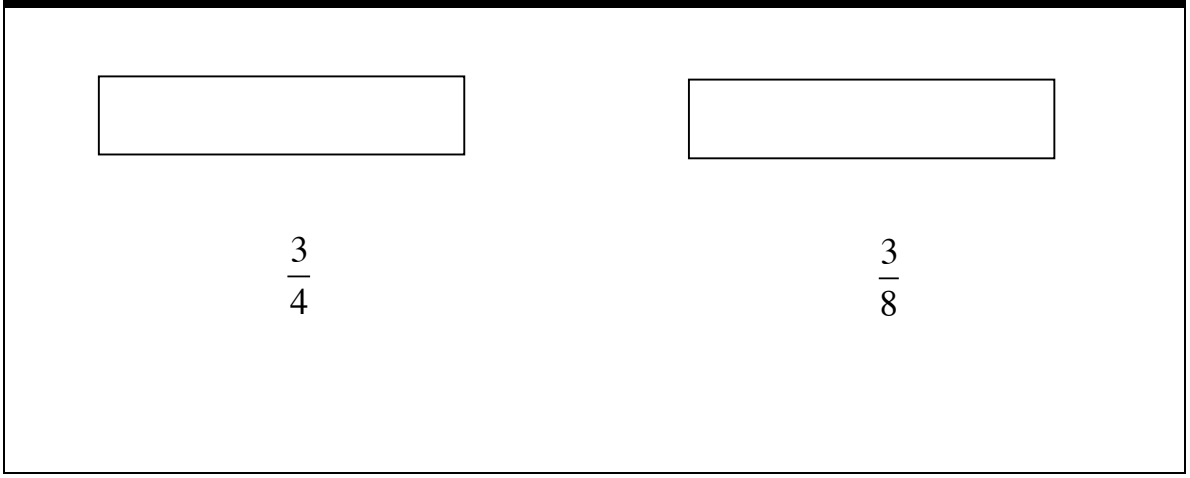
Directions:
Partition and shade each picture to show the fraction. Circle the SMALLER fraction in each pair.



$\frac{1}{2}$ $\frac{1}{4}$



$\frac{1}{3}$ $\frac{1}{6}$



$\frac{3}{4}$ $\frac{3}{8}$

Directions

A friend has been out of school for two days and missed the math lessons dealing with comparing fractions. Write your friend a letter explaining how to compare fractions like the ones you have been working with. [You may want to draw pictures.]